



Calculation Sheet

Calculation header

Identifier

*New calculation (1)***Medium selection and state**

Medium

9% Ammonia

State

*Liquid***Operating data**☐ Safety-related application

	Maximum flow	Mean flow	Minimum flow	
t1	<i>25.0</i>	<i>20.0</i>	<i>10.0</i>	°C
p1	<i>4.08</i>	<i>4.08</i>	<i>4.08</i>	bar(g)
☒ p2	<i>3.01</i>	<i>2.77</i>	<i>2.43</i>	bar(g)
☐ Δp	<i>1.07</i>	<i>1.31</i>	<i>1.65</i>	bar
☒ qm	<i>30.0</i>	<i>16.0</i>	<i>3.0</i>	kg/h

Fluid operating data

	Maximum flow	Mean flow	Minimum flow	
ρ1	☒ <i>0.0</i>	☒ <i>0.0</i>	☒ <i>0.0</i>	kg/m ³
☒ η1	☒ <i>0.0</i>	☒ <i>0.0</i>	☒ <i>0.0</i>	mPa s
cF1	<i>0.0</i>	<i>0.0</i>	<i>0.0</i>	m/s

Pipe downstream of valve

Size class downstream of valve

NPS2

1/2"

Schedule downstream

SCH2

*STD***Valve configuration**

Valve type

Straight globe valve

Trim type

Cage trim

Flow direction

FTO

Valve performance class

Heavy duty valve

Protection

Hardened seat/plug

Low-noise design

n.a.



Calculation Sheet

Valve data

Basic characteristic

Linear☐ Lift stopper

Selected valve size

NPS

1/2"

Pressure class

class

class 300

Maximum operating pressure

p,max

53.0

bar(a)

Valve modifier (Cv/Cv100 = 0.75)

FL²*0.626*

-

Valve modifier (Cv/Cv100 = 0.75)

Kc

0.495

-

Cavitation modifier (Cv/Cv100 = 0.75)

xFz

0.351

-

Valve style modifier (Cv/Cv100 = 1)

Fd

0.41

-

Load-dependent values**Noise calculation**

Calculation standard (noise, liquid)

*IEC 60534-8-4 (2015)***Noise prediction data**

Low-noise design

n.a.

Sound velocity

cF1

0.0

m/s

Legend

Errors

